AIS – AURELIA INDUSTRIAL SYSTEMS

AIS-200 TECHNICAL SPECIFICATIONS

AIS-200 Technical Specifications
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1. Document Purpose

This document defines the technical specifications and engineering characteristics of the AIS-200 Industrial Assembly System.

It provides technical reference information for installation, operation, integration, diagnostics, maintenance, and technical support.

The document is intended for customers, system integrators, maintenance personnel, technical support engineers, and AIS Engineering.

This document complements the AIS-200 User Manual and does not replace applicable safety documentation.

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2. System Overview

The AIS-200 is an industrial automated assembly system designed for medium-to-high throughput production environments.

The system integrates:

- automated assembly stations;
- robotic automation;
- programmable industrial control;

- advanced operator HMI;
- remote diagnostic capabilities;
- redundant safety controls;
- industrial communication interfaces;
- production monitoring.

The AIS-200 is designed for continuous industrial operation where higher throughput and advanced diagnostic capabilities are required.

2.1 Product Identification

Parameter	Specification
Product	AIS-200
Machine type	Industrial Assembly System
Nominal maximum capacity	80 units/min
Control system	AIS Control PLC-200
HMI	AIS Advanced Industrial HMI
Primary communication	Industrial Ethernet
Additional communication	OPC UA
I/O	Digital I/O
Safety architecture	Redundant safety controller
Preventive maintenance interval	Every 2,000 operating hours

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3. Mechanical Specifications

3.1 Dimensions

The standard AIS-200 configuration has the following approximate dimensions:

Parameter	Value

Length	3,200 mm
Width	1,800 mm
Height	2,300 mm
Approximate weight	3,900 kg

Actual dimensions may vary depending on machine configuration, tooling, and optional modules.

Adequate clearance must be provided around the system for operation, maintenance, inspection, and emergency access.

3.2 Mechanical Configuration

The AIS-200 includes:

- main structural frame;
- automated assembly stations;
- robotic modules;
- material handling equipment;
- electrical cabinet;
- cooling-related equipment;
- operator HMI;
- safety guarding and sensors.

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4. Electrical Specifications

The standard electrical configuration is:

Parameter	Specification
Supply voltage	400 VAC
Supply type	Three-phase
Frequency	50/60 Hz

Control voltage	24 VDC
Main control system	AIS Control PLC-200

The electrical installation must comply with the approved machine installation documentation.

Only qualified personnel may perform electrical installation or maintenance.

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5. Performance and Capacity

The AIS-200 has a nominal maximum production capacity of:

80 units per minute

Actual throughput depends on:

- product characteristics;
- production recipe;
- material availability;
- tooling;
- robot cycle times;
- machine configuration;
- upstream and downstream equipment;
- environmental conditions.

The nominal maximum capacity is a technical reference and does not represent a guaranteed production rate for all applications.

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6. Environmental Requirements

The standard operating environment is:

Parameter	Requirement
Operating temperature	5–35 °C
Relative humidity	Up to 80%, non-condensing

Installation environment	Indoor industrial environment
Altitude	Up to 1,000 m without special configuration

The machine must be protected from environmental conditions that may negatively affect electrical, mechanical, robotic, or cooling-system components.

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7. Control System

The AIS-200 uses the **AIS Control PLC-200** as its primary industrial control system.

The control system manages:

- production sequencing;
- machine states;
- robotic coordination;
- sensor inputs;
- actuator control;
- alarm generation;
- HMI communication;
- diagnostic information;
- communication with external systems.

Control software and configuration parameters are machine-specific.

Unauthorized modification of PLC programs or protected configuration parameters is prohibited.

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8. Safety System

The AIS-200 incorporates a redundant safety architecture.

Safety-related components include:

- redundant safety controller;
- emergency stop circuits;

- guard interlocks;
- safety sensors;
- monitored machine states.

Safety systems are designed to place the machine in a defined safe state when a relevant safety condition is detected.

Safety protections must never be bypassed or disabled.

Changes to safety configuration require an approved engineering procedure and qualified personnel.

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9. Cooling System

The AIS-200 includes a cooling system used to maintain operating temperatures within configured limits.

Cooling-system status is monitored by the machine control system.

9.1 Error E-204

Error code E-204 indicates that the cooling-system temperature has exceeded the configured threshold.

When E-204 is reported:

1. Stop the machine safely.
2. Do not continue production while the alarm condition remains unresolved.
3. Check the cooling-system status.
4. Check the coolant level where applicable.
5. Check for visible leakage or abnormal conditions.
6. Verify the cooling unit status.
7. Follow the approved recovery procedure.
8. Contact AIS Technical Support if the alarm persists.

Operators must not:

- bypass the temperature protection;

- disable the alarm;
- modify the configured temperature threshold;
- continue production solely to avoid downtime;
- open equipment requiring qualified intervention.

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10. Communication Interfaces

The standard communication interfaces include:

- Industrial Ethernet;
- OPC UA;
- digital I/O.

Industrial Ethernet provides machine-level network communication.

OPC UA may be used for communication with compatible industrial monitoring or integration systems.

Digital I/O can be used for supported machine-level interfaces.

Network configuration depends on the customer installation and approved machine configuration.

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11. HMI and Diagnostics

The AIS Advanced Industrial HMI provides:

- machine status;
- production information;
- alarm information;
- diagnostic information;
- maintenance information;
- cooling-system status;
- operating parameters;
- selected machine configuration information.

The diagnostic interface can assist operators and technical personnel in identifying machine conditions.

HMI information should be interpreted together with the applicable technical documentation.

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12. Remote Diagnostics

The AIS-200 supports remote diagnostic functionality where the appropriate configuration and customer connectivity are available.

Remote diagnostics may provide authorized AIS personnel with machine status and diagnostic information.

Remote diagnostic access is subject to:

- customer authorization;
- network configuration;
- security controls;
- applicable support procedures.

Remote diagnostic capability does not remove the requirement for local safety procedures.

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13. Utilities and Auxiliary Requirements

The primary utility requirement is the electrical supply specified in this document.

Additional utilities may be required depending on the machine configuration and installed equipment.

Where pneumatic or auxiliary cooling equipment is installed, project-specific requirements are defined in the approved installation documentation.

Customers must verify utility requirements before installation and commissioning.

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14. Maintenance Requirements

Preventive maintenance for the AIS-200 is scheduled every:

2,000 operating hours

Maintenance activities may include:

- mechanical inspection;
- robotic system inspection;
- sensor inspection;
- safety-system verification;
- cooling-system inspection;
- electrical inspection;
- cleaning;
- functional testing.

The maintenance interval specified in the current maintenance documentation is authoritative.

Maintenance history and current machine operating hours should be reviewed when determining whether maintenance is due.

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15. Installation Considerations

Installation requires:

- suitable floor capacity;
- adequate machine clearance;
- electrical supply;
- grounding;
- suitable environmental conditions;
- network connectivity where required;
- access for maintenance;
- appropriate upstream and downstream interfaces.

Because of its size and weight, installation planning must include appropriate lifting and positioning procedures.

Final requirements depend on the supplied configuration.

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16. Configuration and Variants

The AIS-200 can be configured according to customer production requirements.

Configuration differences may include:

- robotic modules;
- tooling;
- cooling equipment;
- sensor configuration;
- communication interfaces;
- HMI configuration;
- production recipes;
- remote diagnostic configuration.

Machine-specific configuration documentation takes precedence over generic technical information.

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17. Technical Support Data

When contacting AIS Technical Support, provide:

- machine model;
- machine serial number;
- operating hours;
- HMI error code;
- description of the issue;
- production impact;
- cooling-system status;
- recent maintenance activities;
- relevant diagnostic information.

For E-204 specifically, provide the time of occurrence, machine state, cooling-system observations, and any recent maintenance or intervention.

18. Document Control

Version	Date	Description
2.0	2025-01-15	Technical specification revision
2.5	2025-07-10	Updated diagnostic and communication information
3.0	2026-01-20	Current technical specification

Current approved version: 3.0